

Geometry

Unit 1

Lesson 10 Homework

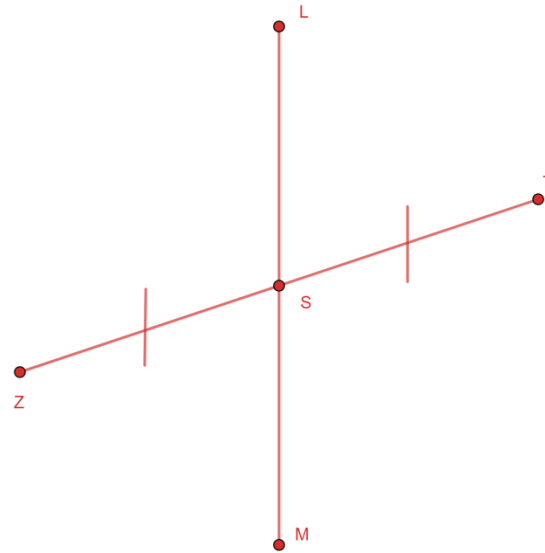
Name **Answer Key** _____

Date _____

Period _____

$\overline{LM}$ bisects $\overline{ZT}$ at point S . $\overline{ZS}$ has a length of $3x + 7$, and $\overline{ST}$ has a length of $-5x + 23$.

1. Draw a diagram of the description.



2. Find the value of x .

$$3x + 7 = -5x + 23$$

$$8x = 16$$

$$x = 2$$

3. Determine the length of $\overline{ZT}$.

$$\overline{ZT} = 3x + 7 + -5x + 23$$

$$= -2x + 30$$

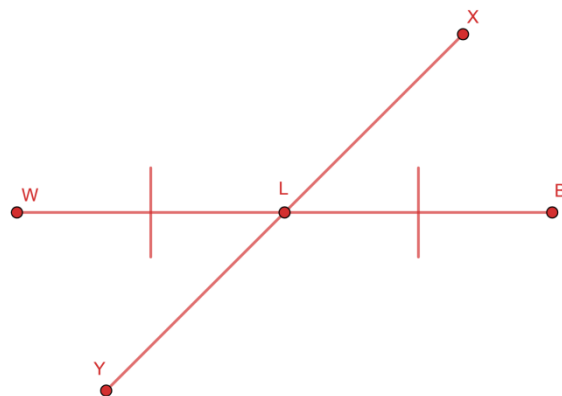
$$= -2(2) + 30$$

$$= -4 + 30$$

$$= 26$$

$\overline{XY}$ bisects $\overline{WB}$ at point L . $WB = 3x + 8$ and $WL = \frac{1}{2}x + 12$.

4. Draw a diagram of the description.



5. Find the value of x .

$$3x + 8 = 2\left(\frac{1}{2}x + 12\right)$$

$$3x + 8 = x + 24$$

$$2x = 16$$

$$x = 8$$

6. Find the length of $\overline{WB}$.

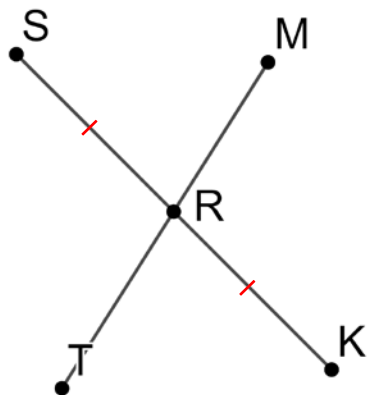
$$\overline{WB} = 3x + 8$$

$$= 3(8) + 8$$

$$= 24 + 8$$

$$= 32$$

$\overline{MT}$ bisects $\overline{SK}$ at point R as shown. $\overline{SK}$ has a length of $24x - 12$, $\overline{RK}$ has a length of $10x - 2$, $\overline{TR}$ has a length of $4x + 5$, and $\overline{RM}$ has a length of $8x - 5$.



7. Based on the given information which two segments are congruent?

$$\overline{SR} \cong \overline{RK}$$

8. Find the value of x .

$$24x - 12 = 2(10x - 2)$$

$$24x - 12 = 20x - 4$$

$$4x = 8$$

$$x = 2$$

9. Find the length of $\overline{RK}$.

$$\overline{RK} = 10x - 2$$

$$= 10(2) - 2$$

$$= 20 - 2$$

$$= 18$$

10. Find the length of $\overline{MT}$.

$$\overline{MT} = 4x + 5 + 8x - 5$$

$$= 12x$$

$$= 12(2)$$

$$= 24$$